Computer Networks | Unit 3

Expert Notes: Network Layer, IPv4, & Routing Algorithms

1. Network Layer Design Issues

Definition

The Network Layer (Layer 3) is the primary layer responsible for end-to-end communication, focusing on moving packets from the source host to the destination host across multiple networks.

Detailed Explanation

The main design philosophy of the network layer is to provide a transparent transfer of data between transport layers. It must know about the topology of the communication subnet and choose appropriate paths through it. One major design issue is the **Store-and-Forward Packet Switching**, where a host sends a packet to the nearest router, which stores it, verifies the checksum, and then forwards it to the next router.
Another critical issue is the choice between **Connection-Oriented (Virtual Circuits)** and **Connectionless (Datagram)** services. In Datagram services, each packet is treated independently and may take different routes. In Virtual Circuits, a fixed path is established before data transfer begins. Additionally, the network layer must handle **Congestion Control** when too many packets are present in the subnet, degrading performance.

Major Design Issues:

- **Routing:** Determining how packets are routed from source to destination.
- **Service Provider Interface:** The network layer must provide services to the transport layer that are independent of router technology.
- **Logical Addressing:** Assigning unique IP addresses to every device globally.
- **Fragmentation & Reassembly:** Breaking large packets into smaller ones if the physical link has a small MTU (Maximum Transmission Unit).

> 👉 ***Hinglish Explanation:*** *Network Layer ka main kaam hai packet ko source se destination tak pahunchana. Ismein sabse bade challenges hain sahi rasta (routing) chunna, traffic (congestion) handle karna, aur badi files ko chhote packets (fragmentation) mein todna.*

2. Internet Protocol Version 4 (IPv4)

Definition

IPv4 is the fourth version of the Internet Protocol. It is a connectionless protocol used in packet-switched layered networks and uses a 32-bit logical address space.

Detailed Explanation

An IPv4 address is written in dotted-decimal notation (e.g., 192.168.1.1). The address consists of two parts: the **Network ID** and the **Host ID**. Because it is 32-bit, the total number of possible addresses is 2^{32} (approx. 4.3 billion), which led to the eventual exhaustion of addresses and the rise of IPv6.
The IPv4 header is typically 20 bytes long. It includes fields like **TTL (Time to Live)**, which prevents packets from looping infinitely by decrementing the count at each router. It also includes the **Checksum** for error detection and **Protocol** field to identify the upper-layer protocol (TCP or UDP).

IPv4 Header Structure (Simplified):

Version/IHL
Type of Service
Total Length
Identification
Flags
Fragment Offset
Time to Live
Protocol
Header Checksum
Source IP Address
Destination IP Address

> 👉 ***Hinglish Explanation:*** *IPv4 32-bit ka address hota hai jo hum devices ko dete hain. Iske header mein "TTL" bahut important hota hai jo packet ko raste mein "phasne" (infinite loop) se bachata hai.*

3. Routing Principles & Algorithms

Definition

Routing is the process of selecting paths in a network along which to send network traffic. A routing algorithm is that part of the network layer software responsible for deciding which output line an incoming packet should go on.

Detailed Explanation

1. Distance Vector Routing (DVR): Each router maintains a table (vector) giving the best-known distance to each destination and which line to use to get there. Routers share their entire tables with their **neighbors only** at regular intervals. *Example: RIP (Routing Information Protocol).* **Problem:** The "Count-to-Infinity" problem occurs when a link goes down, and routers take a long time to realize it.
2. Link State Routing (LSR): Each router discovers its neighbors and learns their network addresses. It then constructs a packet containing all this info (LSP) and **broadcasts it to all routers** in the network. Every router then has a full map of the network and runs **Dijkstra's Algorithm** to find the shortest path. *Example: OSPF (Open Shortest Path First).*

```
[ Distance Vector ] -> Neighbors exchange tables.
[ Link State ]      -> All routers exchange link info (Flood).

Algorithm for LSR (Dijkstra):
1. Create a set of unvisited nodes.
2. Assign to every node a tentative distance value (0 for start, infinity for others).
3. Select node with smallest distance, update neighbors' distances.
```

Routing Protocols:

- **RIP:** Uses hop count as metric (Max 15 hops). Distance vector based.
- **OSPF:** Uses bandwidth/cost as metric. Link-state based. Supports hierarchical routing.

> 👉 ***Hinglish Explanation:*** *Routing matlab best rasta dhoondna. Distance Vector mein router sirf apne padosi se baat karta hai (RIP). Link State mein router pure network ka map banata hai aur Dijkstra algorithm chala kar shortest path nikalta hai (OSPF).*

🔁 Short Revision Notes

- **Network Layer:** End-to-end delivery of packets.
- **IPv4:** 32-bit address, dotted decimal, 20-byte header (standard).
- **Dijkstra's:** Used in Link-state routing to calculate shortest path.
- **RIP vs OSPF:** RIP is for small networks (slow); OSPF is for large/enterprise networks (fast).

- **TTL:** Stops packets from looping forever.

📝 Important Questions (6 Marks)

1. What are the design issues of the Network Layer? Explain Datagram vs Virtual Circuit subnets.
2. Draw and explain the IPv4 header format in detail.
3. Explain the working of Distance Vector Routing. Discuss the Count-to-Infinity problem.
4. Describe the steps of the Link State Routing algorithm with a suitable example.
5. Compare RIP and OSPF routing protocols based on their metrics and performance.

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